CSE4078 Introduction to NLP Spring 2023

FINAL EXAM

PLAN

- **Session 1**
  - Start 15:00
  - End 15:50
  - Upload to ues.marmara.edu.tr in 10 minutes by 16:00
- **Session 2**
  - Start 16:00
  - End 16:45
  - Upload to ues.marmara.edu.tr in 10 minutes by 16:55
- Each question is 25 points.

Ethical Code

- You need to write your full name, your number, class code and date – time on the top of all of your answer papers.
- In the first page, after writing your name and number please write the following statement and sign: ***"I am aware that any forms of cheating in this exam will result in a zero grade and a disciplinary investigation. I accept all rules and regulations regarding online exams. I give permission for the processing of my personal data as stated in the Clarification Text provided on the Faculty of Engineering website"***

SESSION 1

- Start **15:10**
- End **16:00**
- Upload to ues.marmara.edu.tr in 10 minutes by **16:10**

1) A logistic regression model is trained and weight vector w is [-3, -1.1, 0.2, -0.8, 0.3, -1.2, 0.3, -0.3, 4.3, 1, 0.1] for vocabulary [pathetic, worst, part, boxing, scenes, no, plot, twists, great, satire, film] and b=0.1. Classify given documents using this model (p>0.5 is positive). Show your calculations.

D1-“it was pathetic the worst part was boxing scenes”

D2- “it is no pathetic, actually it is great”

D3- “no plot twists or great scenes”

2) Explain key differences between the attention mechanism in LSTM and Transformers (self-attention).

Q1)V=["pathetic", "worst", "part", "boxing", "scenes", "no", "plot", "twists", "great", "satire", "film"]

w = [-3, -1.1, 0.2, -0.8, 0.3, -1.2, 0.3, -0.3, 4.3, 1, 0.1] b=0.1

D1: "it was pathetic the worst part was boxing scenes" **x = [1,1,1,1,1,0,0,0,0,0,0]**

y = **(-3 * 1) + (-1.1 * 1) + (0.2 * 1) + (-0.8 * 1) + (0.3 * 1)** + (-1.2 * 0) + (0.3 * 0) + (-0.3 * 0) + (4.3 * 0) + (1 * 0) + (0.1 * 0) + 0.1

= -3 - 1.1 + 0.2 - 0.8 + 0.3 + 0.1 = -4.3

p = 1 / (1 + e^(-y)) = 1 / (1 + e^(-(-4.3))) ≈ **0.0134** < 0.5 **Negative**.

D2: "it is no pathetic, actually it is great" **x = [1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0]**

y = **(-3 * 1)** + (-1.1 * 0) + (0.2 * 0) + (-0.8 * 0) + (0.3 * 0) + **(-1.2 * 1)** + (0.3 * 0) + (-0.3 * 0) + **(4.3 * 1)** + (1 * 0) + (0.1 * 0) + 0.1 = -3 - 1.2 + 4.3 + 0.1 = 0.2

p = 1 / (1 + e^(-y)) = 1 / (1 + e^(-0.2)) ≈ **0.549** > 0.5 **Positive.**

D3: "no plot twists or great scenes" **x = [0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0]**

Y = (-3 * 0) + (-1.1 * 0) + (0.2 * 0) + (-0.8 * 0) + **(0.3 * 1)** + **(-1.2 * 1)** + **(0.3 * 1)** + **(-0.3 * 1)** + **(4.3 * 1)** + (1 * 0) + (0.1 * 0) +0.1 = 0.3+(-1.2)+0.3+(-0.3)+4.3+0.1 = 3.5

1/(1 + e^-3.5) ≈ 1 / (1 + 0.00316) ≈ 1/1.00316 ≈ **0.997** > 0.5 **Positive**

Q2)

The key differences between the attention mechanism in LSTM and Transformers (self-attention) can be summarized as follows.

- In LSTM, attention operates sequentially and attends to previous time steps, capturing local dependencies within the input sequence.
- It relies on hidden states and recurrent connections.
- On the other hand, Transformers utilize self-attention, which is parallel and attends to all positions in the sequence simultaneously.
- It captures both local and global dependencies efficiently, without the need for recurrent connections or sequential processing.
- Transformers allow for a more comprehensive understanding of context and enable the modeling of long-range dependencies in a parallelized manner.

SESSION 2

- Start **16:15**
- End **17:05**
- Upload to ues.marmara.edu.tr in 15 minutes by **17:20**

	+	*	-
+	50	10	20
*	5	150	5
-	10	10	200

$$\text{score}(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i \cdot \mathbf{x}_j$$

$$\alpha_{ij} = \text{softmax}(\text{score}(\mathbf{x}_i, \mathbf{x}_j)) \ \forall j \le i$$

$$= \frac{\exp(\text{score}(\mathbf{x}_i, \mathbf{x}_j))}{\sum_{k=1}^{i} \exp(\text{score}(\mathbf{x}_i, \mathbf{x}_k))} \ \forall j \le i$$

$$\mathbf{y}_i = \sum_{j \le i} \alpha_{ij} \mathbf{x}_j$$

3) Calculate the Precision, Recall, and F1 scores for each class (A, B, C) and their micro averages of precision and recall.

rows show actual class labels, P=TP/(TP+FP), R=TP/(TP+FN)

4)Assume row vectors represent input vectors x1,x2, x3. Calculate the output y3 of a casual (left-to-right) self-attention layer.

Q3) Precision A = 50 / (50 + 5 + 10) = 0.769

B = 150 / (10 + 150 + 10) = 0.882

C = 200 / (20 + 5 + 200) = 0.888

Recall A = 50 / (50 + 10 + 20) = 0.625

B = 150 / (5 + 150 + 5) = 0.938

C = 200 / (10 + 10 + 200) = 0.909

F1 A = 2 * (50/65 * 50/80) / (50/65 + 50/80) = 0.689

B = 2 * (150/170 * 150/160) / (150/170 + 150/160) = 0.909

C = 2 * (200/225 * 200/220) / (200/225 + 200/220) = 0.898

Total TP = 50 + 150 + 200 = 400

Total FP = 10 + 5 + 10 + 10 + 20 + 5 = 60

Total FN = 10 + 20 + 5 + 5 + 10 + 10 = 60

Micro Avg P = 400 / (400 + 60) = 0.89 = Micro Avg R (because total FP = total FN)

Q4) x1=[50, 10, 20] x2=[5,150,5] x3=[10,10,200]

Score(x1,x3)= 50*5 + 10*150 + 20*5 = 1850

$$\mathbf{y}_i = \sum_{j \le i} \alpha_{ij} \mathbf{x}_j$$

Score(x2,x3)= 5*10 + 150*10 + 5*200 = 2250

Score(x3,x3)= 10*10 + 10*10 + 200*200 = 40200

Softmax(1850, 2250, 40200) can be approximated to [0,0,1] as 40200 is much larger than 2250 and 4600

y3 = 0*[50, 10,20] + 0*[5, 150, 5] + 1*[10, 10, 200]

And similarly y3 can be approximated as x3